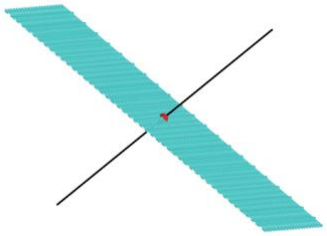
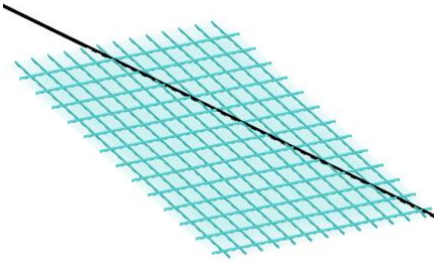
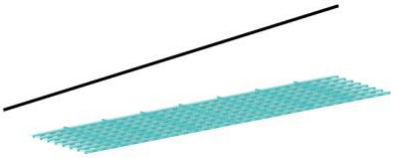
L6 – Intersection of Lines and Planes Unit 6
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Part 1: Intersections of Lines and Planes

Given a line and plane in 3-space, there are three possibilities for the intersection of the line with the plane:

Intersect at 1 Point	Infinite Number of Solutions	No Solutions
	 The line lies on the plane	 The line is parallel and distinct from the plane

Example 1: In each case, determine if the line and the plane intersect. If so, determine the solution.

a) $\pi_1: 9x + 13y - 2z = 29$

$$\ell_1: \begin{cases} x = 5 + 2t \\ y = -5 - 5t \\ z = 2 + 3t \end{cases}$$

Sub the parametric equations in to the scalar equation of the plane:

$9(5 + 2t) + 13(-5 - 5t) - 2(2 + 3t) = 29$

$45 + 18t - 65 - 65t - 4 - 6t = 29$

$-53t = 53$

$t = -1$

Since a single value of t was found, the line and plane intersect at 1 point. Sub $t = -1$ in to the parametric equations:

$x = 5 + 2(-1) = 3$

$y = -5 - 5(-1) = 0$

$z = 2 + 3(-1) = -1$

The line and plane intersect at the point $(3,0,-1)$.

https://www.geogebra.org/3d/a9j3u5u2

b) $\pi_2: x + 3y - 4z = 10$

$$\ell_2: \begin{cases} x = 4 + 6t \\ y = -7 + 2t \\ z = 1 + 3t \end{cases}$$

Sub the parametric equations in to the scalar equation of the plane:

$(4 + 6t) + 3(-7 + 2t) - 4(1 + 3t) = 10$

$4 + 6t - 21 + 6t - 4 - 12t = 10$

$0t = 31$

There are no values of t that can make this equation true. Therefore, the line and the plane do not intersect. The line is parallel and distinct from the plane. https://www.geogebra.org/3d/quagawhg

c) $\pi_3: 4x - y + 11z = -1$

$\ell_3: [x, y, z] = [-2,4,1] + t[3,1,-1]$

Sub the parametric equations in to the scalar equation of the plane:

$4(-2 + 3t) - (4 + t) + 11(1 - t) = -1$

$-8 + 12t - 4 - t + 11 - 11t = -1$

$0t = 0$

This equation is true for all values of t. There are infinitely many solutions. Any point on the line is a solution. The line lies on the plane. https://www.geogebra.org/3d/emxyfbbx

Example 2: Without solving, determine if each line intersects the plane.

a) $\pi_1: 3x - y + z = -6$

$\ell_1: [x, y, z] = [2, -5,3] + s[3,2,1]$

Check if the plane and line are parallel by comparing the direction vector of the line, $\vec{m} = [3,2,1]$ with the normal vector of the plane $\vec{n} = [3, -1,1]$. If they are perpendicular to each other, then the line and plane must be parallel. Find the dot product to analyze:

$\vec{m} \cdot \vec{n} = [3,2,1] \cdot [3, -1,1]$

$\vec{m} \cdot \vec{n} = 3(3) + 2(-1) + 1(1)$

$\vec{m} \cdot \vec{n} = 8$

Since $\vec{m} \cdot \vec{n} \neq 0$, $\vec{m}$ and $\vec{n}$ are not perpendicular. Therefore, the line and plane are NOT parallel and must have one point of intersection.

b) $\pi_2: 4x - 2z = 11$

$\ell_2: [x, y, z] = [1,0,1] + s[-2,1,-4]$

Check if the plane and line are parallel by comparing the direction vector of the line, $\vec{m} = [-2,1,-4]$ with the normal vector of the plane $\vec{n} = [4,0,-2]$. If they are perpendicular to each other, then the line and plane must be parallel. Find the dot product to analyze:

$\vec{m} \cdot \vec{n} = [-2,1,-4] \cdot [4,0,-2]$

$\vec{m} \cdot \vec{n} = -2(4) + 1(0) + (-4)(-2)$

$\vec{m} \cdot \vec{n} = 0$

Since $\vec{m} \cdot \vec{n} = 0$, $\vec{m}$ and $\vec{n}$ ARE perpendicular. Therefore, the line and plane ARE parallel and either have no solutions (parallel and distinct) or infinitely many solutions (line is on plane). Check if the point (1,0,1) is on the plane to determine which situation we have:

LS	RS
$= 4(1) - 2(1)$	$= 11$
$= 2$	

$LS \neq RS$, therefore the line does not lie on the plane. There are no solutions to this system. The line and plane are parallel and distinct.

Part 2: Distance from a Point to a Plane

Remember the formula for the magnitude of the projection of $\vec{a}$ on to $\vec{b}$ is $\left|proj_{\vec{b}}\, \vec{a}\right| = \frac{|\vec{a} \cdot \vec{b}|}{|\vec{b}|}$

The shortest distance between a point and a plane is the perpendicular distance. This distance, d, between a point P and a plane is $d = \left|proj_{\vec{n}}\, \overrightarrow{PQ}\right| = \frac{|\overrightarrow{PQ} \cdot \vec{n}|}{|\vec{n}|}$ where Q is ANY point on the plane and $\vec{n}$ is a normal vector to the plane.

Example 3: Find the distance between the plan $4x + 2y + z - 16 = 0$ and the point $P(10,3,-8)$.

Find any point Q on the plane by choosing arbitrary values for x and y:

$4(1) + 2(1) + z - 16 = 0$

$z = 10$

Point $Q(1,1,10)$ is on the plane.

Vector $\overrightarrow{PQ} = [1,1,10] - [10,3,-8] = [-9,-2,18]$

A normal vector to the plane is $\vec{n} = [4,2,1]$

$$d = \frac{\left|\overrightarrow{PQ} \cdot \vec{n}\right|}{|\vec{n}|}$$

$$d = \frac{|[-9,-2,18] \cdot [4,2,1]|}{\sqrt{(4)^2 + (2)^2 + (1)^2}}$$

$$d = \frac{22}{\sqrt{21}}$$

$d \cong 4.8$ units

https://www.geogebra.org/3d/r5qebzz4